

ZYROO INTERNSHIP PROGRAM

Task 01 — Cybersecurity Virtual Lab Setup

Submission Report

Submitted by

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Lab Summary

Kali Linux was set up as a virtual machine using VMware. Docker was installed inside the Kali VM, and OWASP Juice Shop was deployed as the intentionally vulnerable target application using a Docker container running on port 3000. Connectivity between Kali and the target was verified using curl, which returned valid HTML content from the Juice Shop web application.

1. Docker Installation on Kali

```
(kali@kali)-[~]
└─$ sudo apt install docker.io -y
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
  bluez-hcidump          libfuse2t64          libkmlengine1t64      librubberband2      libxmlsec1-openssl1  python3-aiomcache
  curlftpfs              libgavl-1            libllvmspirvlib18.1   libstdl2-classic     llvm-spirv-18        python3-pluginbase
  enchant-2              libgdal38            libmbedcrypto16       libsframe2          mesa-va-drivers      python3-pysnmp4
  firmware-marvell-presteria libgdk-pixbuf2.0-bin libmupdf25.1          libsodium23          mesa-va-drivers      python3-terminaltables
  libaudio2              libhdf5-310          libntfs-3g89t64       libsphinxbase3t64   network-manager-gnome python3-tz
  libavc1394-0           libhdf5-hl-310       libplacebo351         libteamctl0          openjdk-21-jre        python3-wapiti-arsenic
  libavfilter10          libio-pty-perl       libpocketsphinx3      libteamctl0          openjdk-21-jre-headless python3-yaswfp
  libblsc2-4             libipc-run-perl      libpoppler147         libunibreak6        openssl-client-gssapi  rpcsvc-proto
  libconfig-inifiles-perl libjmc-83            libpostall            libvdpau-va-gl1      pocketchinx-en-us     ruby-unf-ext
  libcrypt-dev           libkmlbase1t64       libpostproc58         libvidstab1.1        postgresql-common-dev python3-aiocache
  libdlt2                libkmlcore1t64       libraw23t64          libxmlsec1-1        python3-aiocache

Use 'sudo apt autoremove' to remove them.

Installing:
  docker.io

Installing dependencies:
  containerd criu docker-buildx docker-cli libcompell python3-protobuf python3-pycrui runc tini-static

Suggested packages:
  containerd-networking-plugins docker-doc btrfs-progs debootstrap rinse rootlesskit xfsprogs zfs-fuse | zfsutils-linux

Recommended packages:
  needrestart

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 201
  Download size: 78.8 MB
  Space needed: 334 MB / 59.0 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 runc amd64 1.3.6+ds1-1 [6,708 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 containerd amd64 2.1.9+ds1-2 [29.5 MB]
Get:3 http://mirror.yer.az/kali kali-rolling/main amd64 libcompell amd64 4.2.1-1 [64.4 kB]
Get:9 http://mirror.ourhost.az/kali kali-rolling/main amd64 python3-protobuf amd64 3.21.12-16 [156 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 criu amd64 4.2.1-1 [557 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 docker-buildx amd64 0.29.1+ds1-4 [14.3 MB]
Get:6 http://http.kali.org/kali kali-rolling/main amd64 docker-cli amd64 28.5.2+dfsg4-4 [8,049 kB]
Get:7 http://http.kali.org/kali kali-rolling/main amd64 tini-static amd64 0.19.0-6+b2 [280 kB]
```

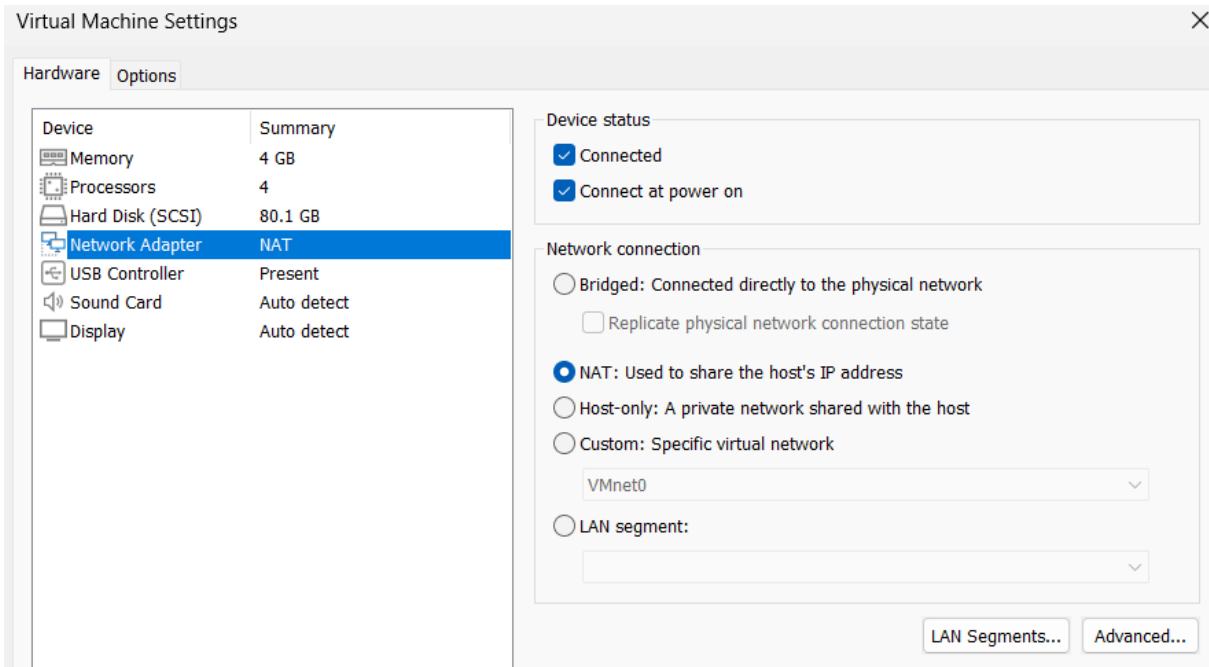
Docker (docker.io) was installed on Kali Linux using apt, providing the runtime used to deploy the OWASP Juice Shop container.

2. Kali Linux IP Address (ip a)

```
(kali@kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:11:27:c4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.43.128/24 brd 192.168.43.255 scope global dynamic noprefixroute eth0
        valid_lft 1722sec preferred_lft 1722sec
    inet6 fe80::36d1:f56e:4324:3408/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 36:46:c7:37:f0:89 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::3446:c7ff:fe37:f089/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
4: veth13c1769aif2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether f2:e5:ba:6f:7a:93 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::f0e5:baff:fe6f:7a93/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

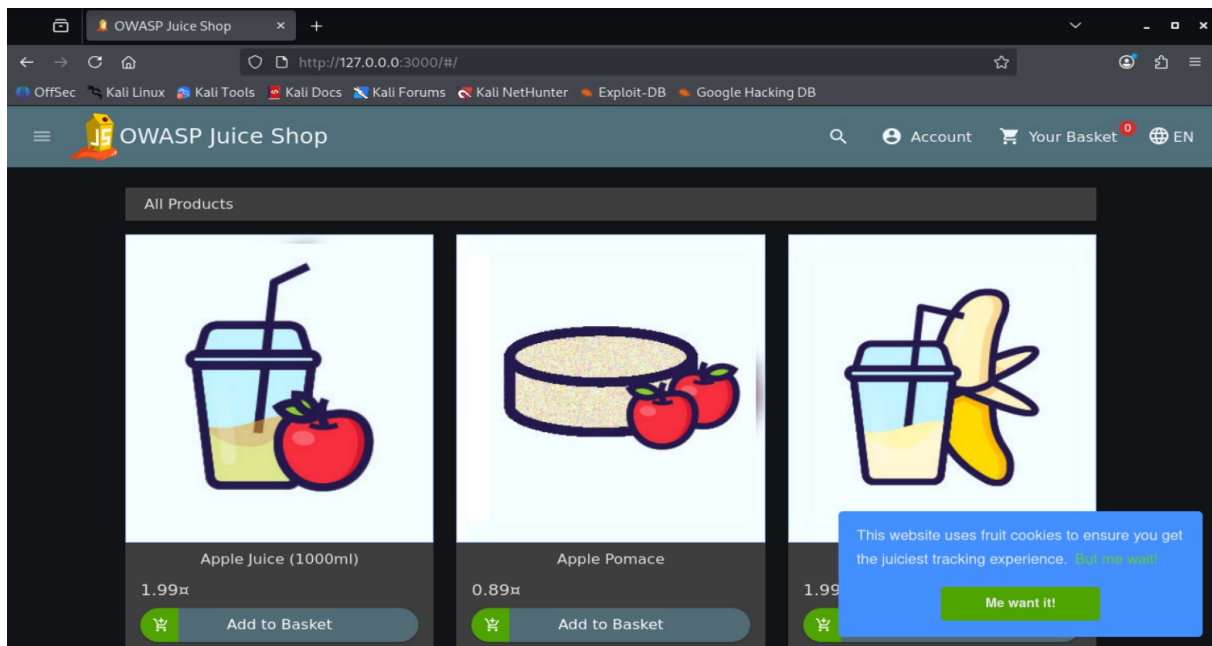
Kali's active network interface (eth0) shows IP address 192.168.43.128. The docker0 interface (172.17.0.1) is used internally by the Juice Shop container.

3. Network Adapter Configuration



VMware Virtual Machine Settings showing the network adapter configuration for the Kali VM.

4. Target Application Running (OWASP Juice Shop)



OWASP Juice Shop loaded successfully in the browser at `http://127.0.0.1:3000`, confirming the target application is running. This also confirms Kali Linux is running and operational.

5. Connectivity Verification (curl)

```
(kali@kali) ~$ curl http://127.0.0.1:8000
<!-- Copyright (c) 2014-2026 Bjoern Kimminich & the OWASP Juice Shop contributors. -->
-- SPOX-License-Identifier: MIT -->

<!doctype html>
<html lang="en" data-beasties-container>
<head>
  <meta charset="utf-8">
  <title>OWASP Juice Shop</title>
  <meta name="description" content="Probably the most modern and sophisticated insecure web application">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <style>font-face{font-family:'VT323';font-style:normal;font-weight:400;font-display:swap;src:url(https://fonts.gstatic.com/s/vt323/v18/pxiKyp0ihIEF2isJFUwOFF.woff2) format('woff2');unicode-range:U+0102-0103,U+0110-0111,U+012B-0129,U+0168-0169,U+01A0-01A1,U+01AF-01B0,U+0300-0301,U+0303-0304,U+0308-0309,U+0323,U+0329,U+1EA0-1EF9,U+20AB;}@font-face{font-family:'VT323';font-style:normal;font-weight:400;font-display:swap;src:url(https://fonts.gstatic.com/s/vt323/v18/pxiKyp0ihIEF2isJFUwOFF.woff2) format('woff2');unicode-range:U+0100-02BA,U+02BD-02C5,U+02C7-02CC,U+02CE-02D7,U+02DD-02FF,U+0304,U+0308,U+0329,U+1D00-1D0F,U+1E00-1E9F,U+1EF2-1EFF,U+20A0-20AB,U+20AD-20C0,U+2113,U+2C60-2C7F,U+A720-A7FE};@font-face{font-family:'VT323';font-style:normal;font-weight:400;font-display:swap;src:url(https://fonts.gstatic.com/s/vt323/v18/pxiKyp0ihIEF2isJFUwOFF.woff2) format('woff2');unicode-range:U+0000-00FF,U+0131,U+0152-0153,U+02BB-02BC,U+02AC,U+02DA,U+02DC,U+0304,U+0308,U+0329,U+2000-206F,U+20AC,U+2122,U+2191,U+2193,U+2212,U+2215,U+FEFF ,U+FFFFD;}</style>
  <link id="favicon" rel="icon" type="image/x-icon" href="/assets/public/favicon.js.ico">
  <script>
    window.addEventListener("load",function(){
      window.cookieconsent.initialise({
        "palette": {
          "popup": { "background": "var(--theme-primary)", "text": "var(--theme-text)" },
          "button": { "background": "var(--theme-accent)", "text": "var(--theme-text)" }
        },
        "theme": "classic",
        "position": "bottom-right",
        "content": { "message": "This website uses fruit cookies to ensure you get the juiciest tracking experience.", "dismiss": "Me want it!", "Link": "B ut me wait!", "href": "https://www.youtube.com/watch?v=9PnKL3wuH4" }
      });
    })
  </script>
  <style>.bluegrey-lightergreen{theme:-mat-sys-background:#121316;-mat-sys-error:#fbbab;-mat-sys-error-container:#28000a;-mat-sys-inverse-on-surface:#2f3033;-mat-sys-inverse-surface:#000000;-mat-sys-inverse-surface:#3e3e6e;-mat-sys-on-background-surface:#3e3e6e;-mat-sys-on-error:#690005;-mat-sys-on-error-contain ner:#ffdad6;-mat-sys-on-primary:#000f65;-mat-sys-on-primary-container:#d73eff;-mat-sys-on-primary-fixed:#001b3f;-mat-sys-on-primary-fixed-variant:#004545;-mat-sys-on-secondary:#283041;-mat-sys-on-secondary-container:#dae2f9;-mat-sys-on-secondary-fixed:#131c2b;-mat-sys-on-secondary-fixed-variant:#3e474f}
  </style>
```

The command 'curl http://127.0.0.1:3000' returns valid HTML content from the Juice Shop application, confirming connectivity between Kali and the target.

Checklist Summary

- Docker installed successfully on Kali Linux
- Kali Linux running successfully
- Target application (OWASP Juice Shop) running successfully via Docker
- Target reachable from Kali on the same machine (isolated, not exposed to public internet)
- Kali IP address recorded (192.168.43.128)
- HTTP connectivity verified with curl
- Required screenshots captured